



Session 3 – Performing Conditional Formatting

Session Objectives

By the end of this session, students will be able to:

- 1. Apply built-in conditional formatting rules to highlight data patterns.
- 2. Create and customize their own conditional formatting rules.
- 3. Import and export XML data for advanced data handling.



1. Understanding Conditional Formatting

Conditional Formatting allows Excel to **automatically apply formatting** (like color, font, or icon sets) to cells that meet specific conditions or rules.

Use Case Example:

In our **Sales_Data** sheet, we can automatically highlight sales with *Total Sales* above ₦10,000 or highlight *top 10 performers*.



2. Built-in Rules to Highlight a Cell or Range

Built-in Conditional Formatting Options:

Go to: **Home** → **Conditional Formatting** → **Highlight Cells Rules / Top/Bottom Rules / Data Bars / Color Scales / Icon Sets**

Common Built-in Rules

Rule Type	Description	Example (Using Sales_Data)
Greater Than / Less Than	Highlights cells greater or less than a value.	Highlight <i>Total_Sales</i> > ₦10,000

Rule Type	Description	Example (Using Sales_Data)
Equal To / Between	Highlights exact matches or ranges.	Highlight <i>Units_Sold</i> between 20 and 30
Text That Contains	Highlights cells with certain text.	Highlight <i>Region</i> that contains “South”
Date Occurring	Highlights cells with dates matching certain periods.	Highlight <i>Date</i> in “Last Month”
Duplicate Values	Highlights duplicates.	Highlight duplicate <i>SalesPerson</i> names
Top/Bottom Rules	Highlights top or bottom numbers or percentages.	Highlight top 10 <i>Total_Sales</i>

Class Demonstration

1. Open **Sales_Data** sheet.
2. Select the **Total_Sales** column.
3. Go to: Home → Conditional Formatting → Highlight Cells Rules → Greater Than...
4. Enter **10000** → Choose a fill color (e.g., Green).
5. Click OK.

Observe that all cells with sales > ₦10,000 are now highlighted automatically.

Real-life Application

- In sales analysis: to instantly identify high-performing products.
- In finance: to track overdue payments or low balances.
- In HR: to highlight employees with performance scores above a target.

◆ 3. Customized Conditional Formatting Rules

Creating Your Own Rule

Steps:

1. Select data range → Home → Conditional Formatting → New Rule .
2. Choose **“Use a formula to determine which cells to format.”**
3. Enter a formula that returns TRUE/FALSE.

Example 1:

Highlight sales where region is “West” and total sales > ₦15,000

```
=AND($B2="West", $G2>15000)
```

Then choose a bold font and light blue fill.

Example 2:

Highlight employees in **Employee_Data** whose performance score ≥ 8:

```
=$F2>=8
```



Practice Exercise

Basic:

Highlight all salespersons with sales over ₦20,000.

Intermediary:

Highlight rows in *Sales_Data* where Region = "East" **AND** Units_Sold > 25.

Advanced:

Use icon sets to represent performance levels in *Employee_Data*:

- Green icon for scores ≥ 8
- Yellow for scores between 5–7

- Red for scores < 5

◆ 4. Managing and Editing Rules

To view or modify existing rules:

1. Select the range → Home → Conditional Formatting → Manage Rules .
2. Choose **“This Worksheet”** from the dropdown.
3. Edit, delete, or reorder rules.

Tip:

If multiple rules apply to the same range, Excel applies them **from top to bottom** (use the “Stop If True” option to prevent overlap).

◆ 5. Importing and Exporting XML Data

XML (Extensible Markup Language) is used to store and transport structured data, often between systems.

Importing XML Data:

1. Go to: Developer → Import → Select XML File .
2. Choose an XML file with matching schema (e.g., product or employee data).
3. Excel will map the XML elements into columns automatically.

Example:

Import employee data structured like:

```
<Employee>
  <Name>Mary</Name>
  <Department>HR</Department>
  <Salary>52000</Salary>
</Employee>
```

Exporting XML Data:

1. Prepare your data table (like **Employee_Data**).
2. Go to: Developer → Export → Choose XML file path.
3. Excel exports structured data for integration with other systems.

Use Case:

- Export employee performance data to share with a web dashboard.
- Import product catalog updates from a company database.



Classwork Summary

Level	Activity
Basic	Apply conditional formatting to highlight sales > ₦10,000 in <i>Sales_Data</i> .
Intermediary	Use a custom rule to highlight employees with performance ≥ 8.
Advanced	Create a 3-color scale for <i>Total_Sales</i> column and export <i>Employee_Data</i> as XML.



Summary

- Conditional formatting visually emphasizes data trends.
- You can use both **built-in** and **custom rules**.
- XML import/export enhances **data interoperability** between Excel and databases or apps.